

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TRANSMISSION CONTROL SWITCH (TCS)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Transmission Control Switch (TCS). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [External Controls](#) (307-05B Automatic

Transmission/Transaxle External Controls - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1142-62	Ignition Status 1 - Signal compare failure	Mismatch between hardwired ignition signal and CAN powermode signal	Refer to the electrical circuit diagrams and check the ignition signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
P0603-44	Internal Control Module Keep Alive Memory (KAM) Error - Data memory failure	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0604-44	Internal Control Module Random Access Memory (RAM) Error - Data memory failure	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0605-45	Internal Control Module Read Only Memory (ROM) Error - Program memory failure	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0606-2F	ECM / PCM Processor - Signal erratic	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0606-47	ECM / PCM Processor - Watchdog / safety microcontroller failure	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0705-09	Transmission Range Sensor A Circuit	Transmission control switch	Using the manufacturer approved diagnostic system,

	(PRNDL Input) - Component failures	failure	clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0705-46	Transmission Range Sensor A Circuit (PRNDL Input) - Calibration/parameter memory failure	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0705-86	Transmission Range Sensor A Circuit (PRNDL Input) - Signal invalid	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P0814-01	Transmission Range Display Circuit - General electrical failure	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P081C-64	Park Input Circuit - Signal plausibility failure	Mismatch between hardwired park signal and transmission control switch gear selection	NOTE: Park signal should be high (battery voltage) when park selected and low (ground) for all other gear selections Refer to the electrical circuit diagrams and check the park signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
P084F-11	Park/Neutral Switch Output Circuit - Circuit short to ground	Park / neutral signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the park/neutral signal circuit for short circuit to ground
P084F-15	Park/Neutral Switch Output Circuit - Circuit short to battery or open	Park / neutral signal circuit short circuit to power, open	Refer to the electrical circuit diagrams and check the park/neutral signal circuit for short circuit to power, open

		circuit, high resistance	circuit, high resistance
P176A-01	Transmission Range Selector Up and Down Position Circuit - General electrical failure	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176A-13	Transmission Range Selector Up and Down Position Circuit - Circuit open	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176A-19	Transmission Range Selector Up and Down Position Circuit - Circuit current above threshold	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176A-94	Transmission Range Selector Up and Down Position Circuit - Unexpected operation	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176B-71	Transmission Range Selector Up and Down Position Control Error - Actuator stuck	- Transmission control switch raise/lower operation has been forced/abused - Level of software in the transmission control switch is incorrect - Transmission control switch internal failure	- Check for evidence of obstruction or abuse and rectify as appropriate - Using the manufacturer approved diagnostic system check and install the latest relevant level of software to the transmission control switch - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176C-07	Transmission Range Selector Lock Control Error - Mechanical	Transmission control switch gear selection	

	failures	operated while locking Transmission control switch failure	NOTE: This DTC may be set if the driver releases the brake pedal while the transmission control switch is between selections - Test the locking / unlocking operation of the transmission control switch - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176C-11	Transmission Range Selector Lock Control Error - Circuit short to ground	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176C-15	Transmission Range Selector Lock Control Error - Circuit short to battery or open	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
P176C-73	Transmission Range Selector Lock Control Error - Actuator stuck closed	- Transmission control switch unlock operation obstructed - Transmission control switch failure	NOTE: This DTC may be set by driver abuse, namely by applying a rotational force before the transmission control switch is unlocked - Test the locking / unlocking operation of the transmission control switch - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new

			transmission control switch
U0001-86	High Speed CAN Communication Bus - Signal invalid	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid signal source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	Missing message from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- Car configuration file mismatch with vehicle specification - Transmission control switch failure	- Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
U1A14-04	CAN Initialization Failure - System internal failures	Transmission control switch is not configured correctly	Using the manufacturer approved diagnostic system, perform routine - Configure existing module - Transmission control switch

		Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch
U2100-00	Initial Configuration Not Complete - No sub type information	Transmission control switch failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control switch